

intermittent recurrences. There is at present no way to predict the course a given patient will take.

Traditionally, treatment of sJIA has begun with nonsteroidal anti-inflammatory medications along with corticosteroids. While some patients do well with this treatment, more than 60% require steroids for more than 6 months, and the likelihood of developing chronic arthritis may not be diminished. Inhibitors of TNF- α (see Case 35) have been used therapeutically in sJIA since the 1990s. Unfortunately, the response rate to these medications is significantly lower than in other types of JIA, and they are largely ineffective for the systemic (fever, rash, organomegaly) manifestations of this disease.

More recently, therapies relying on the inhibition of IL-1 and IL-6 have undergone clinical trials. Increased levels of IL-6 have been observed in the serum and joint fluid of patients with sJIA, and these levels correlate with disease activity. Spikes in IL-6 levels also parallel the fever curve seen in sJIA. Tocilizumab is a humanized monoclonal antibody directed against the IL-6 receptor that has recently been approved by the US Food and Drug Administration (FDA) for the treatment of rheumatoid arthritis and for sJIA, where it is remarkably effective for the treatment of both systemic and arthritic manifestations.

Aberrant IL-1 release has also been shown to have a role in many familial autoinflammatory disorders (see Case 33). Although levels of soluble IL-1 are difficult to measure in sJIA, this limitation may be technical. Serum from patients with sJIA has been shown to induce IL-1 secretion as well as the transcription of various innate immune genes from peripheral blood mononuclear cells from healthy individuals. IL-1 blockade can be highly effective in sJIA. In new-onset febrile disease, utilization of the recombinant IL-1 receptor antagonist anakinra results in steroid-free remission in more than half of the patients, and is therefore a common first-line therapeutic agent. The anti-IL-1 β antibody canakinumab has also shown striking efficacy in randomized controlled trials, and is FDA approved for sJIA as well as for other conditions of IL-1 excess. The optimal therapeutic strategy for utilization of IL-1 versus IL-6 blockade in sJIA remains to be determined.

Questions.

- 1 Why is sJIA considered to be primarily an autoinflammatory syndrome, rather than simply an arthritis?
- 2 What might be some possible side effects of IL-1 or IL-6 inhibition?
- 3 What other conditions involve IL-1 dysregulation?
- 4 Why is Catherine's rash only present in association with fever?

CASE 35

RHEUMATOID ARTHRITIS

A common, severe inflammatory disease of unknown cause.

The adaptive immune system can incite inflammation and cause extensive tissue damage and destruction. The utility of this response to the host is not always apparent, and the result may even be detrimental. Rheumatic disease—inflammation and damage to joints and other tissues—is caused by a subset of autoimmune diseases (Fig. 35.1 and see Case 34). Of these, rheumatoid arthritis is the most common, affecting 0.5–1% of the adult population in the United States. This chronic, debilitating disease is chiefly characterized by inflammation of the synovium, the thin tissue layer lining the joint. As the disease progresses, inflamed synovium invades cartilage and bone, leading to areas of focal injury termed erosions. Local inflammation also recruits neutrophils into the joint fluid, where they produce cytokines and proteolytic enzymes that perpetuate inflammation and also injure the cartilage directly. This process can lead ultimately to joint destruction (Fig. 35.2). Without proper treatment, patients with rheumatoid arthritis suffer chronic pain, loss of function, and long-term disability.

Rheumatic diseases caused by autoimmunity

- Systemic lupus erythematosus (SLE)
- Rheumatoid arthritis
- Juvenile arthritis
- Sjögren's syndrome
- Scleroderma (progressive systemic sclerosis)
- Polymyositis–dermatomyositis
- Behçet's disease
- Ankylosing spondylitis
- Reiter's syndrome
- Psoriatic arthritis

Fig. 35.1 The spectrum of autoimmune rheumatic diseases.

TOPICS BEARING ON THIS CASE:

Inflammatory responses

Cytokines and chemokines

Genetic associations with autoimmune disease

Anti-cytokine therapeutic agents



Fig. 35.2 X-ray of the right hand of a patient with rheumatoid arthritis. It shows extensive destruction and dislocation of the metacarpophalangeal joints. A bone erosion at the proximal interphalangeal joint is indicated by an arrowhead.

Rheumatoid arthritis exhibits a strong association with a specific set of major histocompatibility complex (MHC) class II alleles. Since MHC class II molecules present antigen to CD4 T cells, the connection between the disease and MHC class II molecules strongly implicates CD4 T cells in the pathogenesis of rheumatoid arthritis. Autoantibodies are also prevalent in rheumatoid arthritis, including rheumatoid factor (RF), a term that encompasses antibodies (often IgM, but also other subclasses) that are directed against the Fc portion of IgG. RF is found in some healthy individuals and accompanying other disease states, and is absent from a subset of patients with rheumatoid arthritis, leaving its role in disease pathology uncertain. However, complement-fixing immune complexes are abundant in the joints of most RF-positive patients, consistent with a pathogenic role for these and other autoantibodies. According to the current paradigm, in a genetically susceptible host who encounters an unknown environmental trigger, a break in tolerance results in the generation of autoreactive CD4 T cells. These cells infiltrate the joint to induce inflammation and tissue injury via the elaboration of cytokines and chemokines. T cells also provide help to autoreactive B cells, leading to their differentiation into plasma cells that produce arthritogenic autoantibodies. The relative contribution of T cells and autoantibodies likely varies from patient to patient. Beyond immune elements, the joint stroma likely plays a key role. Synovial fibroblasts provide the environment in which the immune response becomes established within the joint. Further, under the influence of inflammatory cytokines, fibroblasts themselves assume a pro-inflammatory and invasive phenotype, and join osteoclasts and activated chondrocytes as major effectors of structural injury to the joint.

The case of Ariadne Underwood: swollen joints that are stiff and painful in the morning portend a common diagnosis.

Ariadne Underwood teaches a class of sixth-graders. When she was 30 years old, Ariadne developed stiffness in her shoulders, hands, knees, and hips. The stiffness lasted for about an hour every morning, and increased in intensity over the course of several months. After her wrists and fingers began to swell, she saw her primary care physician. Her medical history indicated that a maternal aunt had rheumatoid arthritis, a disease that led to markedly deformed hands, wrist swelling that required surgical fusion for pain control, and bilateral total knee replacements. Based on these findings, the primary care doctor suspected that Ms Underwood had rheumatoid arthritis and advised her to take four aspirin tablets a day. He also obtained a blood sample, which was positive for RF.

The patient was referred to a rheumatologist, Dr Shaw. On physical examination, the patient's blood pressure, pulse, and respiratory rate were all normal, and she had no fever. Dr Shaw felt spongy swelling indicating thickening of the synovium in both elbows, as well as swelling and decreased flexion and extension of her wrists, second and third metacarpophalangeal (MCP) joints, and second and third proximal interphalangeal (PIP) joints. Her hemoglobin was 11.4 g dl^{-1} , and her hematocrit was 33.4% (she was slightly anemic). Her white blood cell count was $4720 \text{ } \mu\text{l}^{-1}$ (slightly low). Her sedimentation rate was 87 mm h^{-1} (very high). A test for recent parvovirus infection was negative, while anti-cyclic citrullinated peptide (CCP) antibodies were positive (see below). A urinalysis was normal. X-rays of her hands and feet revealed erosions in several joints. Dr Shaw confirmed the diagnosis of seropositive, erosive rheumatoid arthritis ("seropositive" indicates positive tests for RF and/or CCP).

Treatment was initiated with ibuprofen, a nonsteroidal anti-inflammatory drug (NSAID), 600 mg every 6 hours; prednisone, a glucocorticoid preparation, 10 mg every morning; and methotrexate, a disease-modifying antirheumatic drug (DMARD), 20 mg weekly. At her follow-up visit one month later, the patient reported substantial

Start on a TNF- α antagonist.
Joint swelling and pain.

Autoantibodies.

relief from pain, but joint swelling and tenderness persisted. Three months after initiation of treatment, joint inflammation (synovitis) was still evident on examination. Dr Shaw initiated treatment with weekly subcutaneous etanercept (Enbrel), a TNF- α antagonist. Ms Underwood experienced rapid improvement in her pain and morning stiffness, and 9 months after diagnosis her joint exam normalized and she was able to discontinue NSAIDs and steroids altogether. Over the next several years, she was able to continue all her usual activities, and serial X-rays revealed no further joint erosion, although she experienced occasional disease flares managed with NSAIDs or short courses of prednisone. She was unable to taper off either methotrexate or etanercept.

Rheumatoid arthritis.

Rheumatoid arthritis is an inflammatory, destructive polyarthritis that characteristically affects the small joints of the hands and feet, most typically including the wrists and second and third MCPs and PIPs, though inflammation and injury in larger joints is also common. Epidemiologically, the disease affects females three times more commonly than males. Many patients present in early adulthood, though the disease may appear as early as late childhood and incidence rises gradually throughout adult life. There is no single diagnostic test for rheumatoid arthritis. RF is positive in 60–70% of patients, although as noted this marker is not specific for rheumatoid arthritis. Antibodies against citrullinated proteins (measured as anti-cyclic citrullinated peptide, CCP) are commonly found in patients with rheumatoid arthritis, typically patients who are also positive for RF. Although these antibodies are highly specific for rheumatoid arthritis, 30–40% of patients lack anti-CCP antibodies, and other disease characteristics allow diagnosis even in the absence of positive serologic assays (Fig. 35.3). Ariadne Underwood's presentation was typical, and her positive RF and CCP tests leave no doubt about her diagnosis, though in seronegative young women a parvovirus B19-mediated transient arthritis can present similarly.

As noted, there is a strong association between certain MHC class II alleles (including HLA-DRB1*0404 and DRB1*0401) and seropositive rheumatoid arthritis. This association is driven by four specific amino acids in the antigen-presenting groove that favor the binding of peptides that are citrullinated—that is, that have been post-translationally modified by enzymes called peptidyl arginine deiminases (PADs) to convert positively charged arginine to the neutral amino acid citrulline. Citrullination is a necessary step in the generation of neutrophil extracellular traps (NETs) and other physiologic processes. Patients with seropositive rheumatoid arthritis commonly exhibit reactivity to citrullinated peptides derived from multiple proteins. A leading hypothesis for the origin of rheumatoid arthritis is that initial exposure to citrullinated peptides occurs after these are generated in the lungs as a result of exposure to cigarette smoke (the leading environmental risk factor for rheumatoid arthritis) or in the mouth through the action of

Joint erosions by X-ray.

Criteria for the diagnosis of rheumatoid arthritis

- Morning stiffness lasting at least 1 hour before maximal improvement
- Arthritis of three or more joints simultaneously with swelling and/or fluid in the joints
- Arthritis in hand joints with swelling in the wrists, or metacarpophalangeal joints or proximal interphalangeal joints
- Symmetrical arthritis of the same joint areas
- Rheumatoid nodules
- Serum rheumatoid factor
- Typical radiographic changes

Fig. 35.3 Key features of rheumatoid arthritis according to the 2010 American College of Rheumatology/ European League Against Rheumatism classification criteria (https://www.rheumatology.org/practice/clinical/classification/ra/ra_2010.asp).

Porphyromonas gingivalis, a bacterial contributor to gingivitis that also expresses PADs and is therefore enzymatically capable of citrullination. These hypotheses may help to explain the intriguing observation that rheumatoid arthritis was apparently unknown in Europe before the discovery of the New World, which became Europe's source of both tobacco and sugar. Murine data suggest that anti-citrullinated peptide antibodies (ACPAs) become pathogenic where incidental joint inflammation (e.g., from a viral or traumatic cause) results in citrullination within the joint.

Genome-wide association studies have identified associations between rheumatoid arthritis and over 100 genetic loci beyond the MHC, including many encoding proteins important for immune system function such as *PTPN22*, *STAT4*, *TRAF1*, *CTLA4*, *IL2*, *IL21*, and *CD40*. Some of these "hits" implicate pathways targeted by effective antirheumatic therapies, e.g., the T cell costimulatory blocker CTLA4 (mimicked by CTLA4-Ig, abatacept) and the IL-6 pathway (targeted by the IL-6 receptor blocker tocilizumab). This suggests that the remainder may point to additional pathways of therapeutic relevance.

Animal models have shed considerable light on the pathogenesis of rheumatoid arthritis. Multiple models are available. In most models, arthritis arises through deposition of IgG autoantibodies in joints, either because the target antigen resides there (e.g., anti-collagen antibody arthritis) or because the target becomes deposited upon the negatively-charged cartilage surface (e.g., K/BxN arthritis). Other arthritis models are driven directly by CD4 T cells, independent of autoantibodies (e.g., SKG arthritis, IL-1ra-deficient arthritis). These models have identified pathogenic roles for certain lineages (neutrophils, fibroblasts, mast cells, platelets) as well as for specific molecules including complement and a range of cytokines, chemokines, Fc receptors, and adhesion molecules (Fig. 35.4). Extrapolation of these findings to human disease is useful but must be performed with caution, both because mice and humans are separated by 75 million years of evolutionary time and because each model reflects only a subset of the processes likely operative in human rheumatoid arthritis. However, murine experiments paved the way for the introduction of TNF- α inhibitors into clinical practice, and these model animals continue to represent a valuable tool to understand arthritis biology *in vivo*.

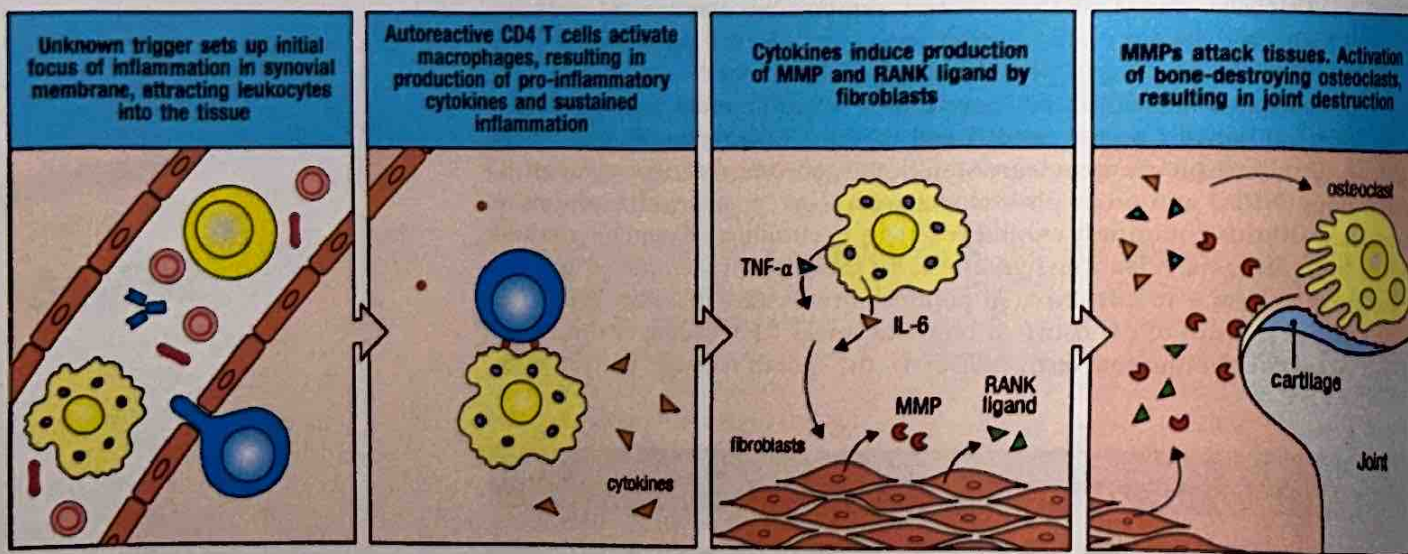


Fig. 35.4 The pathogenesis of rheumatoid arthritis. Inflammation of the synovial membrane, initiated by some unknown trigger, attracts autoreactive lymphocytes and macrophages to the inflamed tissue. Autoreactive effector CD4 T cells activate macrophages, with the production of pro-inflammatory cytokines such as IL-1, IL-6, IL-17, and TNF- α . Fibroblasts activated by cytokines produce matrix

metalloproteinases (MMPs), which contribute to tissue destruction. The TNF-family cytokine RANK ligand, expressed by T cells and fibroblasts in the inflamed joint, is the primary activator of bone-destroying osteoclasts. Antibodies against several joint proteins are also produced (not shown), but their role in pathogenesis is uncertain.

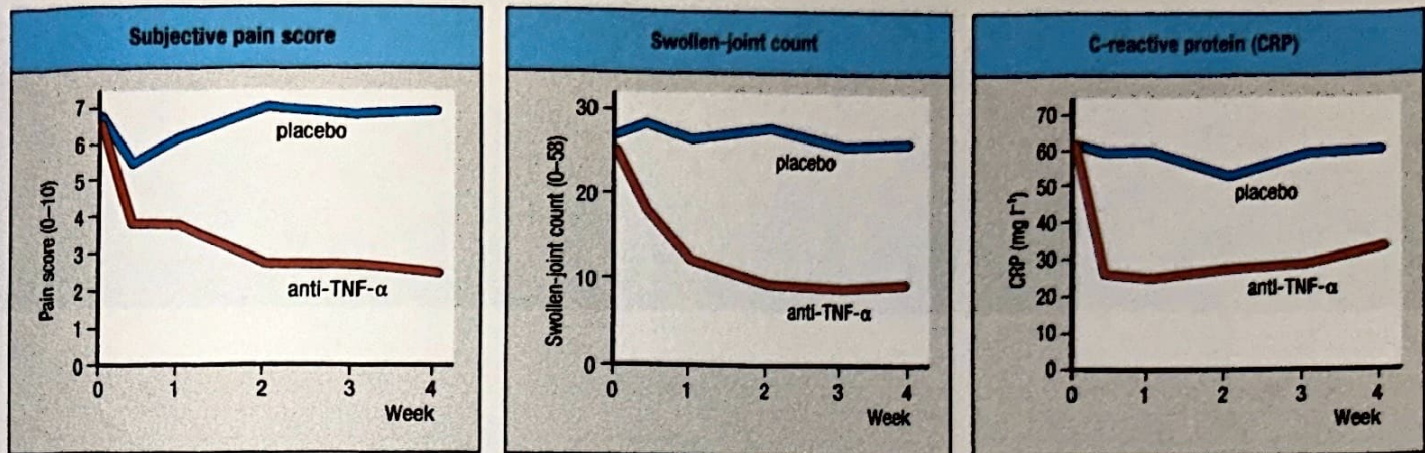


Fig. 35.5 The effects of treatment of rheumatoid arthritis with TNF- α antagonism. For each parameter measured (pain, swollen joints, and level of C-reactive protein), the values for patients given a placebo treatment are depicted by the blue curve and the values for patients treated with anti-TNF- α antibody are depicted by the red curve.

In the large majority of patients, rheumatoid arthritis is a lifelong illness requiring ongoing medical treatment. Fortunately, a growing number of therapeutic options has reduced the debilitating problems resulting from disabling joint injury (Fig. 35.5). NSAIDs provide limited symptomatic relief, but do not alter the course of disease. Glucocorticoid treatment can serve as a helpful therapeutic bridge, controlling swelling and pain until other drugs have a chance to work, though effective doses are typically too toxic for long-term use. In most cases, disease-modifying therapy begins with methotrexate, a folate antagonist that also increases tissue adenosine, an anti-inflammatory mediator. Methotrexate may be combined with sulfasalazine (an antibiotic derivative with an undefined mechanism of action) and hydroxychloroquine (an anti-malarial drug that appears to antagonize intracellular Toll-like receptors) as “triple therapy.”

More recently, an improved understanding of rheumatoid arthritis has led to the development of biological disease-modifying drugs. The most prevalent of these are the TNF- α inhibitors. The first of these to be used, etanercept, is a recombinant protein consisting of a pair of TNF- α receptors (TNFR2) taking the place of the Fab regions in an IgG1 molecule, thereby trapping secreted TNF- α . Anti-TNF monoclonal antibodies such as infliximab and adalimumab function similarly, but have a slightly different effector profile because they bind surface TNF- α as well. Other effective “biologicals” include abatacept, a CTLA4-Ig fusion protein that blocks T cell costimulation via CD28; tocilizumab, an antibody that blocks the IL-6 receptor; rituximab, an antibody against CD20 that depletes B cells; and tofacitinib, an orally available small molecule that antagonizes JAK3 and JAK1, signaling kinases that mediate the action of cytokines including IL-6. Anakinra, the recombinant IL-1 receptor antagonist, is approved for rheumatoid arthritis but has marginal efficacy and is rarely used for this indication (but see Case 34 for its utility in other inflammatory disorders). Together, these agents have revolutionized the care of rheumatoid arthritis and changed the prognosis for newly diagnosed patients, albeit at great expense—medications often cost more than \$50,000 per year.

Questions.

- 1 The presence of rheumatoid factor in the serum of a patient is not diagnostic of rheumatoid arthritis. Why is this?
- 2 How do TNF- α and IL-1 enhance the infiltration of leukocytes into the joint space?
- 3 What might be the risks of TNF- α inhibitor therapy?